

$$f_X(x) = \lim_{\delta \rightarrow 0} \frac{P(X \leq x+\delta) - P(X \leq x)}{\delta}$$

$$f_X(x) = \frac{\alpha x_m^\alpha}{x^{\alpha+1}} \quad x \geq x_m$$

$$= 0 \quad \text{otherwise}$$

$$E(X) = \int_{x_m}^{\infty} x \frac{\alpha x_m^\alpha}{x^{\alpha+1}} dx$$

$$= \alpha x_m^\alpha \left[\frac{x^{-\alpha+1}}{-\alpha+1} \right]_{x_m}^{\infty} \quad \text{undefined for } \alpha < 1.$$